

Education

H.B.Sc., University of Toronto (Astronomy & Astrophysics, Statistics, Mathematics) 2018-2022

Publications

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2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. The Mass of the Milky Way from the H3 Survey, submitted to ApJ.
 1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. 2021, ApJ, 917, 79. [2105.11572](#).

Research Experience

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| 2021 May - 2021 Aug | Undergraduate Summer Research Fellowship
<i>Canadian Institute for Theoretical Astrophysics</i>
Supervisors: Prof. Gwendolyn Eadie , Prof. Norman Murray , Dr. Josh Speagle
Continued previous work on the mass distribution of the Milky Way. Collaborated with members of the H3 Survey and published a first-author paper in ApJ. Began a project using normalizing flows to make probabilistic predictions of the ages of stars and a project using Stan to fit dynamical models to geological data from thermal tides. |
| 2020 Aug - 2021 May | Undergraduate Research Course
<i>Canadian Institute for Theoretical Astrophysics</i>
Supervisors: Prof. Gwendolyn Eadie , Prof. Norman Murray
Developed a hierarchical Bayesian model for modelling the mass of galactic halos using a fast and scalable sampling algorithm. |
| 2020 May - 2020 Aug | Summer Undergraduate Research Program
<i>Dunlap Institute for Astronomy and Astrophysics, University of Toronto</i>
Supervisor: Prof. Allison Man
Analyzed ALMA data to infer molecular gas masses in several high-redshift galaxies. Resulted in a first-author publication. |

Presentations

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| 2021 August | SDSS 2021 Collaboration Meeting, JHU/Online
Lightning talk: "Predicting the ages of stars with machine learning." |
| 2021 May | Stellar Stats Workshop, UofT/Online
Lightning talk: "Estimating the mass distribution of the Milky Way with Bayesian multilevel models." |
| 2021 May | Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online
Poster presentation and lightning talk: "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$." |
| 2021 Apr | Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online
Talk: "Molecular gas in gravitationally lensed galaxies at $z = 2.9$." |
| 2020 Oct | REQUIEM Galaxies Telecon, Online
Talk: "Molecular gas in gravitationally lensed galaxies at $z = 2.9$." |
| 2020 Aug | Summer Undergraduate Research Program Poster Session, UofT/Online
Poster presentation: "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$."
Awarded prize for one of top three posters. |

Honours and Awards

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| 2021 May - 2021 Aug | Undergraduate Summer Research Fellowship, (\$8,396 CAD)
Canadian Institute for Theoretical Astrophysics, University of Toronto |
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Selected Projects

- 2021 Comprehensive scientific computing package for Rust: SIMD-accelerated linear algebra, statistical distributions, numerical optimization, linear models, and more. github.com/al-jshen/compute
- 2021 Zero-dependency reverse-mode automatic differentiation package. github.com/al-jshen/reverse
- 2021 Fast and easy random number generation in Rust with Wyrand. github.com/al-jshen/alea
- 2020 Parallelized ray tracer for realistic scene rendering. github.com/al-jshen/traycer

Media

- 2021 Jul **National Radio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: “Molecular gas in high redshift galaxies”, **Jeff Shen** and Allison Man.
https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas

Outreach

- 2019 - Present **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - present.

Volunteering

- 2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

Technical Skills

Data: Stan, Pyro, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, SQL, R

Other: Rust, BLAS/LAPACK, SIMD intrinsics, Astropy, L^AT_EX, regex, bash, git, Docker, AWS, Javascript, React